

# GradBoost — Model Evaluation

| field                          | value                      |
|--------------------------------|----------------------------|
| model_label                    | GradBoost                  |
| estimator_class                | GradientBoostingClassifier |
| param.ccp_alpha                | 0.0                        |
| param.criterion                | friedman_mse               |
| param.init                     | None                       |
| param.learning_rate            | 0.1                        |
| param.loss                     | log_loss                   |
| param.max_depth                | 3                          |
| param.max_features             | None                       |
| param.max_leaf_nodes           | None                       |
| param.min_impurity_decrease    | 0.0                        |
| param.min_samples_leaf         | 1                          |
| param.min_samples_split        | 2                          |
| param.min_weight_fraction_leaf | 0.0                        |
| param.n_estimators             | 200                        |
| param.n_iter_no_change         | None                       |
| param.random_state             | 0                          |
| param.subsample                | 1.0                        |
| param.tol                      | 0.0001                     |
| param.validation_fraction      | 0.1                        |
| param.verbose                  | 0                          |
| param.warm_start               | False                      |

## GradBoost — SUMMARY OUTPUT (Classification)

| Metric            | Value                    |
|-------------------|--------------------------|
| Test observations | 2348                     |
| Accuracy          | 0.5268                   |
| Balanced accuracy | 0.4716                   |
| Macro precision   | 0.4893                   |
| Macro recall      | 0.4716                   |
| Macro F1          | 0.4396                   |
| Weighted F1       | 0.4828                   |
| Dominant error    | 0 (Loss) → 2 (Win) [278] |

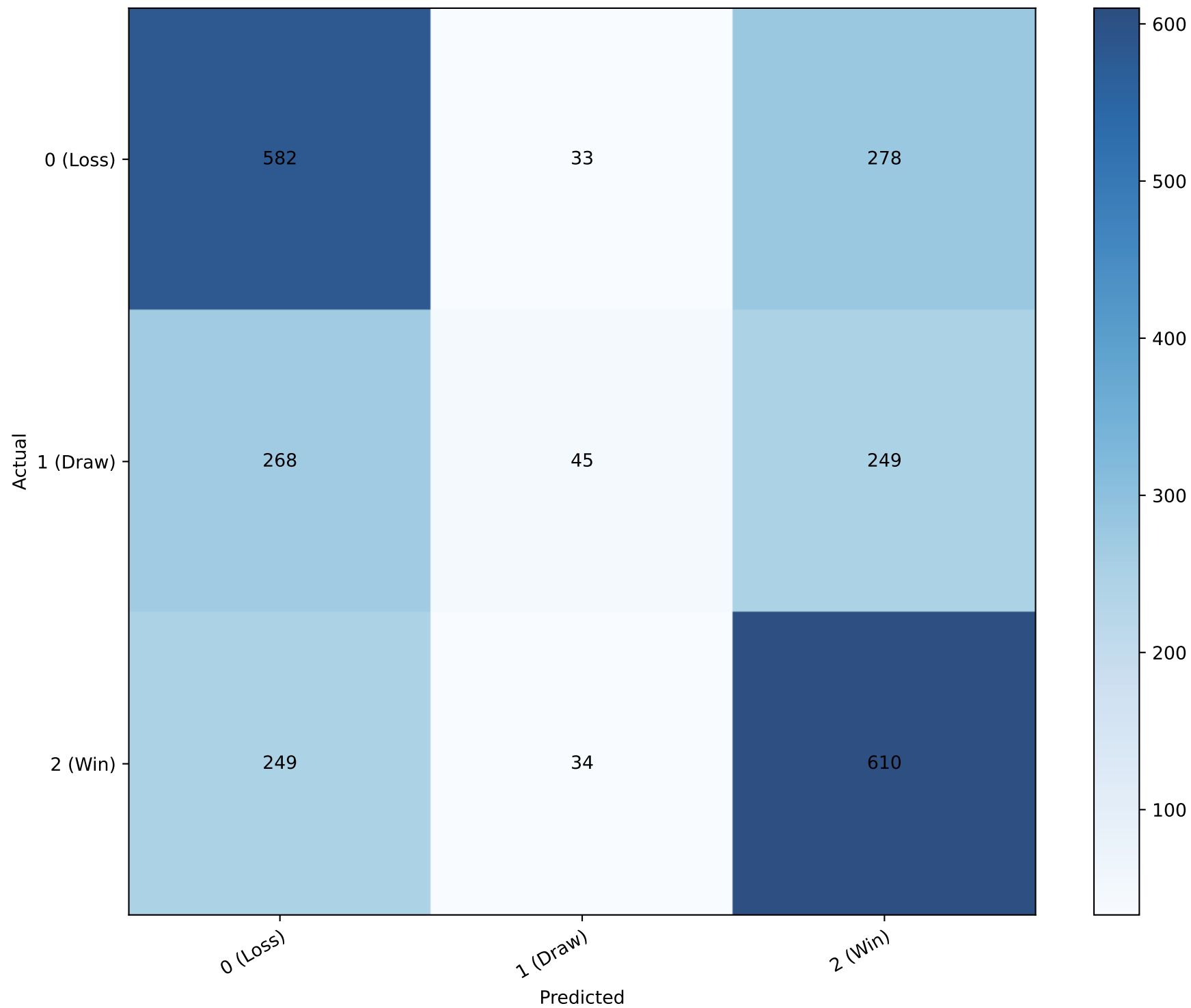
GradBoost — Metrics Summary

| metric             | value               |
|--------------------|---------------------|
| accuracy           | 0.5268313458262351  |
| balanced_accuracy  | 0.47163253404959354 |
| precision_macro    | 0.48928587100717075 |
| recall_macro       | 0.47163253404959354 |
| f1_macro           | 0.43961790944749896 |
| precision_weighted | 0.5016208590288072  |
| recall_weighted    | 0.5268313458262351  |
| f1_weighted        | 0.48276727698188676 |
| n_test             | 2348.0              |

GradBoost — Classification Report

| class_or_avg | precision           | recall              | f1-score            | support            |
|--------------|---------------------|---------------------|---------------------|--------------------|
| 0            | 0.5295723384895359  | 0.6517357222844344  | 0.5843373493975904  | 893.0              |
| 1            | 0.4017857142857143  | 0.0800711743772242  | 0.13353115727002968 | 562.0              |
| 2            | 0.5364995602462621  | 0.683090705487122   | 0.6009852216748769  | 893.0              |
| accuracy     | 0.5268313458262351  | 0.5268313458262351  | 0.5268313458262351  | 0.5268313458262351 |
| macro avg    | 0.48928587100717075 | 0.47163253404959354 | 0.43961790944749896 | 2348.0             |
| weighted avg | 0.5016208590288072  | 0.5268313458262351  | 0.48276727698188676 | 2348.0             |

GradBoost — Confusion Matrix



## GradBoost — Feature Importances (Top 13)

| feature            | importance           |
|--------------------|----------------------|
| b365_team_win_prob | 0.5548904885785666   |
| b365_draw_prob     | 0.11441493356103866  |
| xG_last3           | 0.056046089197700766 |
| xGA_last3          | 0.04815215132406244  |
| xGA_last5          | 0.04730897776465109  |
| xG_last5           | 0.046196541124765844 |
| b365_away_prob     | 0.03979908740961161  |
| b365_home_prob     | 0.03831845608120832  |
| pts_last5          | 0.016899041697926    |
| B365A              | 0.010424550148942415 |
| B365H              | 0.010008093120425153 |
| B365D              | 0.009211694557691283 |
| pts_last3          | 0.00832989543341006  |